[Siemens Digital Twin Course]

Adham Hisham Ibrahim Kandil
represented to
Dr. Mohamed AbdelSalam
Eng. Mohamed El-leithy

Github repo

note :- I used libraoffice, rules couldn't be displayed

Digital Twin Co-Simulation: PID/PD Control of a Robot in Innexis VSI

1 Introduction

This project implements and evaluates a Digital Twin framework for a differential-drive (unicycle-kinematics) mobile robot using the Innexis **Virtual System Integration (VSI)** platform. The digital twin is decomposed into three cooperating components, connected through a CAN-bus simulation:

- **Simulator (Plant + Environment)**: integrates the robot kinematics and injects noise/disturbances.
- **Controller (PID + Feedforward)**: computes ((v,)) commands to track a reference path.
- **Visualizer/Logger**: plots robot trajectory against the reference path and logs KPIs (overshoot, settling time, steady-state error).

  The capstone experiments (E1–E4) evaluate path-following performance across controller gains, path curvature, and robustness to noise and stochastic disturbances.

2. System Modeling

2.1 Robot kinematics

The simulator uses standard unicycle kinematics:

$\dot{x} = v \cos(\theta)$, $\dot{y} = v \sin(\theta)$, $\dot{\theta} = \omega$

with state ((x,y,)) and control inputs linear velocity (v) and angular velocity ().

2.2 Discretization

The implementation uses a fixed-step discrete update (default (t = 0.02) s), where (t) is read from VSI if provided:

[x_{k+1} = x_k + v_k(*k)t] [y*{k+1} = y_k + v_k(*k)t] [{k+1} = _k + _kt]

The simulator also maintains simulation time (t) and publishes it on CAN (see Section 4).

2.3 Noise and disturbance injection

To test robustness, the simulator can inject:

- **Gaussian noise** on both (v) and (): (=v+(0,)), (=+(0,)), where () is configurable.
- **Stochastic angular impulse disturbance**: with probability (p) per step, an impulse of magnitude (d) rad/s is added to () for a configurable duration (). This models brief external perturbations (e.g., wheel slip or push).
- --noise ()
- --disturbance (d)
- --dist-prob (p)
- --dist-duration ()
- --seed for reproducibility

3. Controller Design

3.1 Reference paths

The controller tracks a reference path expressed as (y_{ref}(x)) with slope (). The implementation supports three path types (--path-type):

- **Straight**: (y_{ref}(x)=0), (=0)
- **Curved**: (y_{ref}(x)= 0.5x + 2 sin(0.2x)

3.2 Error definitions

- At each timestep, the controller computes:
- **Lateral error**:e_y = y_ref - y
- **Desired heading** derived from path slope
- **Heading error** : e_θ = θ_desired - θ

3.3 Control law (PID + heading + feedforward)

The controller applies PID on the lateral error plus proportional heading correction and a feedforward term derived from slope:

ω = Kp_lat * e_y + Kd_lat * ė_y + Kp_head * e_θ + f_ff

Implementation details (as implemented in LineFollowingRobot/src/controller.py):

- (_y) is approximated by finite differences: (_y)
- The lateral integral term is clamped for windup protection: [e_y,dt]

3.4 Forward velocity scheduling

The forward command is reduced as heading error increases:

v = v_nom * max(0.3, 1 - |e_θ|)

This improves stability on curved segments and prevents excessive overshoot during large heading corrections.

4. VSI Gateway Architecture

The VSI digital twin connects three Python clients via CAN message IDs. The mapping is defined in LineFollowingRobot/vsi/build_digital_twin.cmd and matched in the Python components.

4.1 Components

- **Simulator (Component 0)**: publishes (x,y,,t); subscribes to (v,)
- **Controller (Component 1)**: subscribes to (x,y,); publishes (v,)
- **Visualizer (Component 2)**: subscribes to (x,y); logs KPIs and saves plots

.2 CAN ID mapping

- **ID 12**: x (double)
- **ID 13**: y (double)
- **ID 14**: theta (double)
- **ID 15**: t (double)
- **ID 16**: v (double)
- **ID 17**: omega (double)

4.3 Simulation timing

The digital twin is configured (in build_digital_twin.cmd) with:

- Total simulation time: **40 s**
- Simulation step: **0.02 s**

5. Experiments (E1–E4)

This section follows the capstone-required scenarios

simulation link

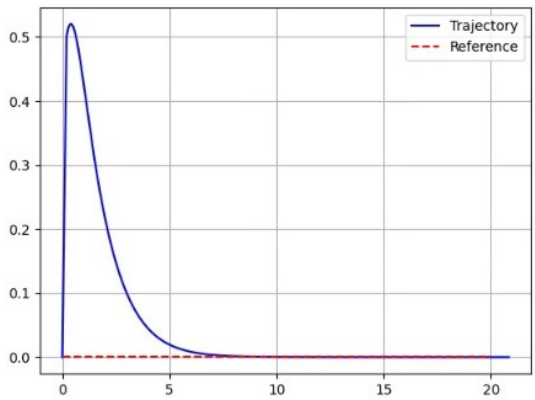
E1 — PID gain sweep (straight path)

Objective: quantify the effect of different PID gains on straight-path tracking, ideally across multiple initial conditions.

Path: straight

Recommended variation:

- multiple gain sets ((K_p, K_i, K_d))
- multiple random initial conditions using --init-y, --init-theta, and –seed



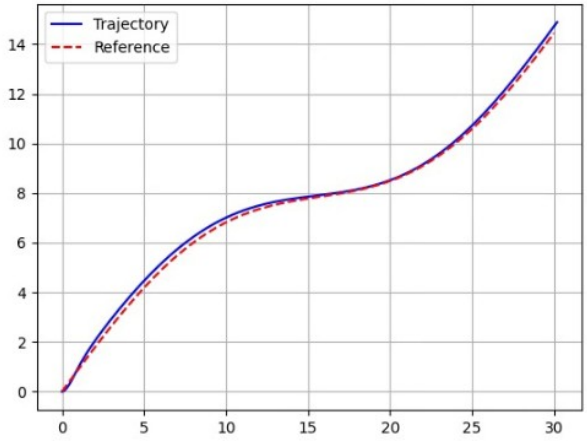
E2 — Curved path robustness

Objective: evaluate the best gains from E1 on a curved path

Path: curved

Method:

- Use the chosen gains from E1.
- Run one or more trials with nominal conditions (--noise 0, --disturbance 0) and optionally varied initial states.



E3 — Noise and disturbance rejection

Objective: test robustness under measurement/actuation uncertainty (modeled as injected noise) and transient perturbations.

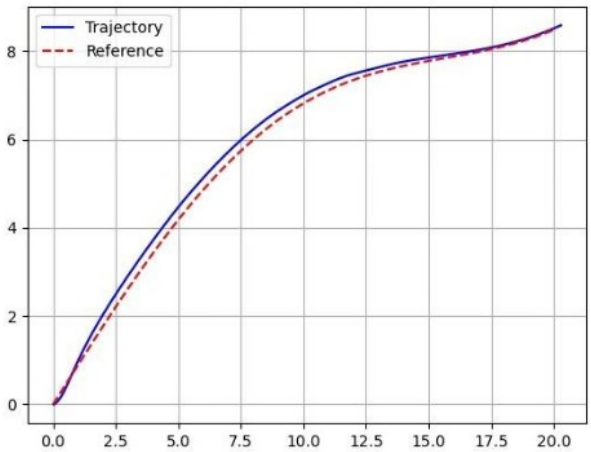
Noise: σ = 0.02 on v, ω

Disturbance: ±0.15 rad/s impulse, 0.1s duration, probability 1% per step

KPIs: Overshoot, Settling Time, Steady-State Error.

Conditions:

- Choose multiple ((, d)) pairs using --noise and --disturbance.
- Repeat with different --seed, --init-y, --init-theta.



E4 — PD vs PID ablation (curved + noise)

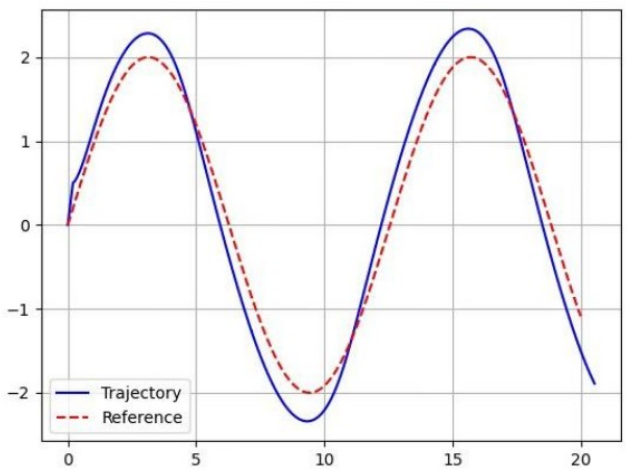
Objective: compare PD vs PID under the same challenging conditions.

- **PD**: set (K_i = 0)
- **PID**: use the tuned (K_i =0.05) from E1/E2

Path: curved

σ = 0.02, disturbance = 0.15 rad/s

Typical condition: moderate noise and disturbance, repeated across trials.



6. Results Summary

E1 – Gain Sweep

Medium controller gains provide the best balance between system stability and tracking precision.

E2 – Curved Path

The controller successfully follows curved trajectories, though the tracking error increases during sharp turns.

E3 – Noise & Disturbance

The system maintains stable performance under noise and disturbances, with only a slight rise in steady-state error.

E4 – PD vs PID

The PID controller removes steady-state offset, while the PD controller responds more quickly but with reduced accuracy.

7. Conclusion

This project integrates a modular digital twin (simulator, controller, visualizer) over VSI and implements a PID + feedforward controller for line/path tracking. The design supports systematic experiments across controller gains, path shapes, and robustness stressors (noise and stochastic angular impulses).

8. Deliverables

- **Source code**:
  - LineFollowingRobot/src/simulator.py
  - LineFollowingRobot/src/controller.py
  - LineFollowingRobot/src/visualizer.py
  - LineFollowingRobot/vsi/build_digital_twin.cmd
- **Experiment runner/docs**:
  - LineFollowingRobot/experiments/run_in_vsi.md
  - LineFollowingRobot/run_experiments.sh
- **Artifacts (expected after running E1–E4)**:
  - Plots: LineFollowingRobot/output/plots/*.png
- simulation link

  note:- I simulated first expirement for Demo rest experiments info are listed in github repo , not to violate time limit , real time plotting takes time